

6.115 Final Project Report

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1 Introduction

Literature loves secret passageways, from worlds hidden within wardrobes to magic walls that can open with a touch. Taking inspiration from the famous scene in Harry Potter where a brick wall opens up to reveal Diagon Alley, this project aims to develop a small version of its own unique puzzle wall, where the user will need to press “bricks,” enter proper numerical combinations, and answer riddles in order to make the wall open up.

2 Hardware System Design

The primary hardware involved in this project is a PSoC 5LP CY8CKIT-050 controlling four main systems through GPIO: a TFT display, a speaker, a ‘wall,’ and a capacitive touch sensor.

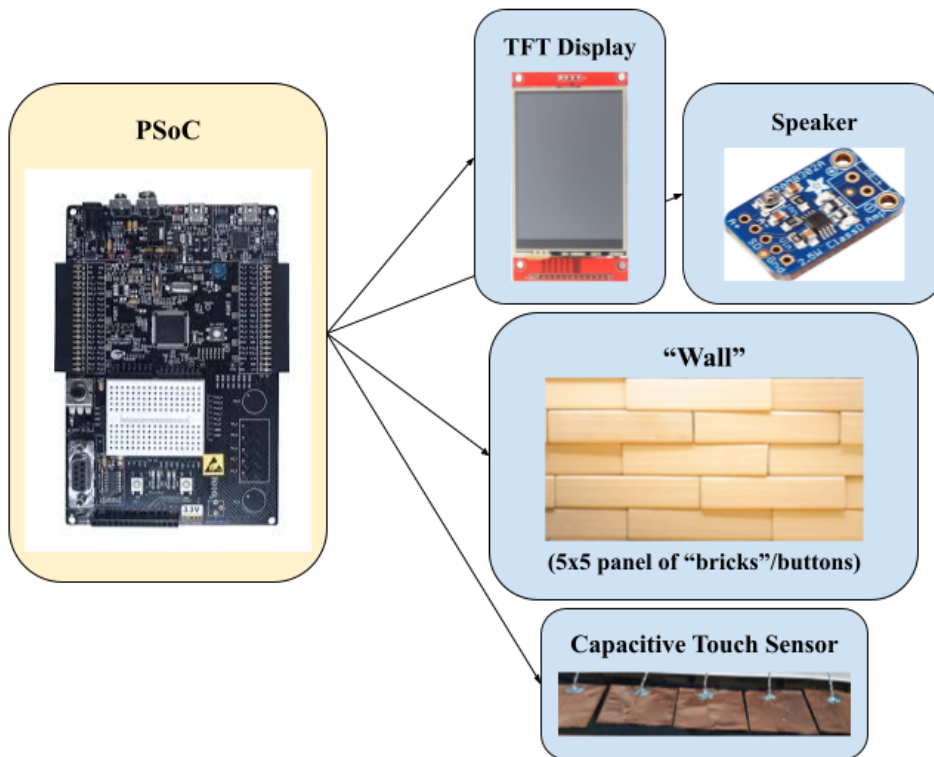


Figure 1: A block diagram of the hardware system design.

The ILI9341 TFT display is used to prompt the user and react to user input.

The PAM8302A audio amplifier is connected to a speaker in order to provide additional auditory feedback.

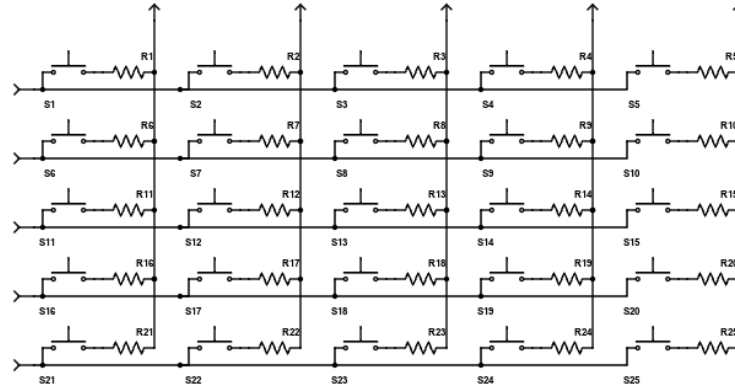


Figure 2: Schematic of the 5x5 button matrix.

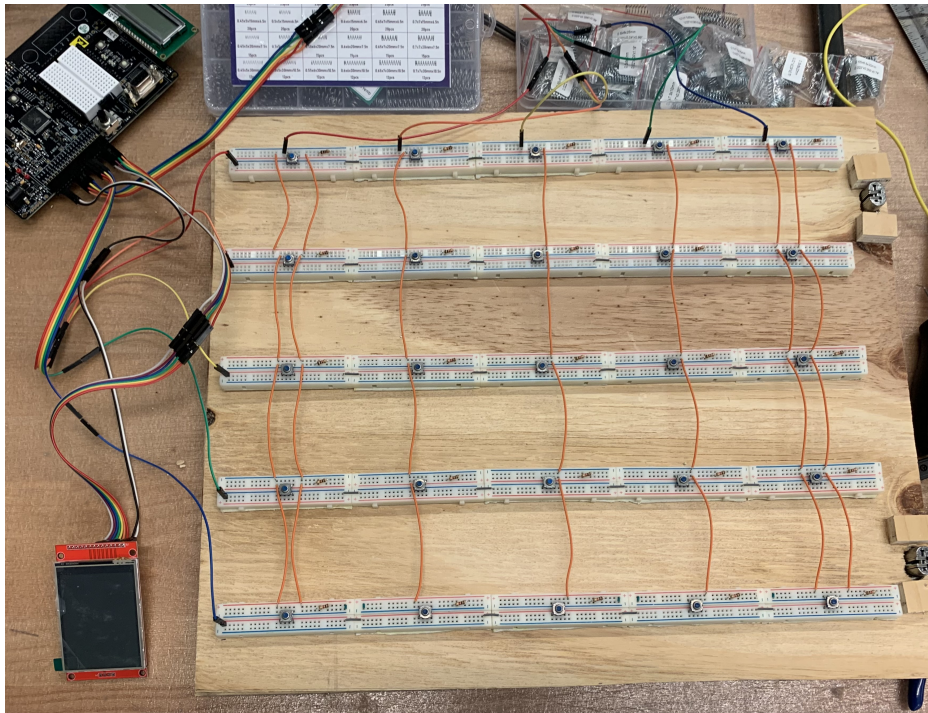


Figure 3: The internal wiring of the 5x5 button panel used for the wall.

The 'wall' consists of a 5x5 panel of twenty-five 'bricks' (i.e. 3.5"x3.5" wooden blocks) mounted on buttons. These buttons are connected in a 5x5 matrix to the PSoc GPIO pins, acting as user input.

The capacitive touch sensor is a panel with strips of copper tape connected to the PSoc GPIO pins. These touch sensor plates act as an additional source of user input, taking advantage of the human body's electrical properties to detect when a copper strip is pressed and respond accordingly.

For the hardware construction, the wall itself consists of two backboards connected together by a hidden, 180 degree hinge. The panel of buttons/bricks is mounted to the top backboard, where each brick is attached to springs around the buttons to provide an improved tactile response. Future iterations of this project will mount the electric lock onto the backboards, allowing the wall to be 'opened' (i.e. the two backboards to be separated) only when the lock is properly triggered.

3 Software System Design

At a high level, the software involved in this project can be modeled as a state machine with four core states: 'Waiting,' 'Name,' 'Code,' and 'Unlocked.'

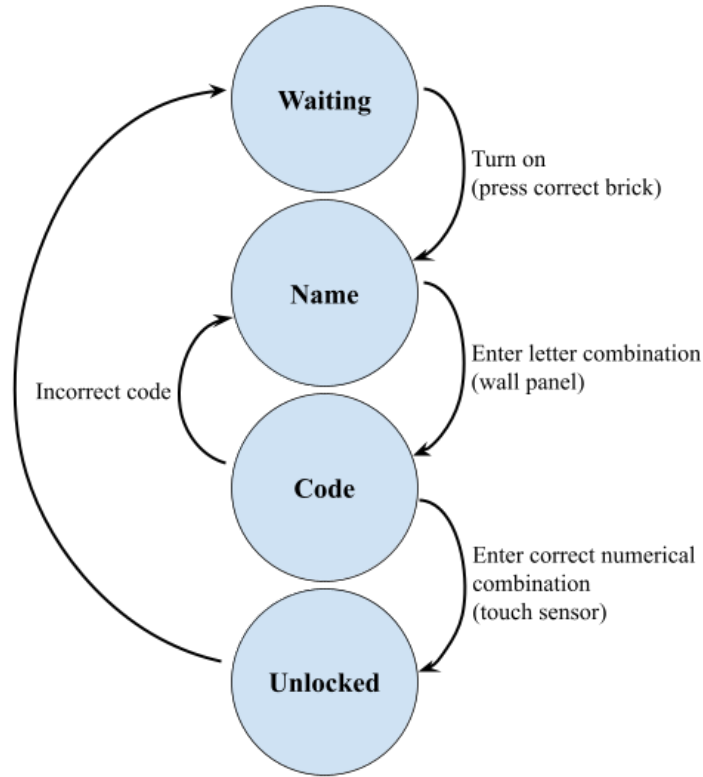


Figure 4: A high-level state machine diagram of the software system design.

In the Waiting state, the system waits for the correct brick to be pressed in order to begin (in this case, it is the brick that is three up and two across from the bottom-right corner, in the style of Harry Potter).

Once the correct brick is pressed, the system enters the Name state, offering the user the option to enter their name by pressing corresponding bricks (where each brick corresponds to a letter of the alphabet). Certain special names will cause unique responses (e.g. "SIMON" will allow the user to play a game of Simon Says, "SNAKE" will allow the user to play a game of Snake, "CODE" will allow the user to enter their code manually using bricks, "RESET" will allow the user to reset their code, and "OFF" will bring the system back to the Waiting state), but any normal response will bring the user to the Code state.

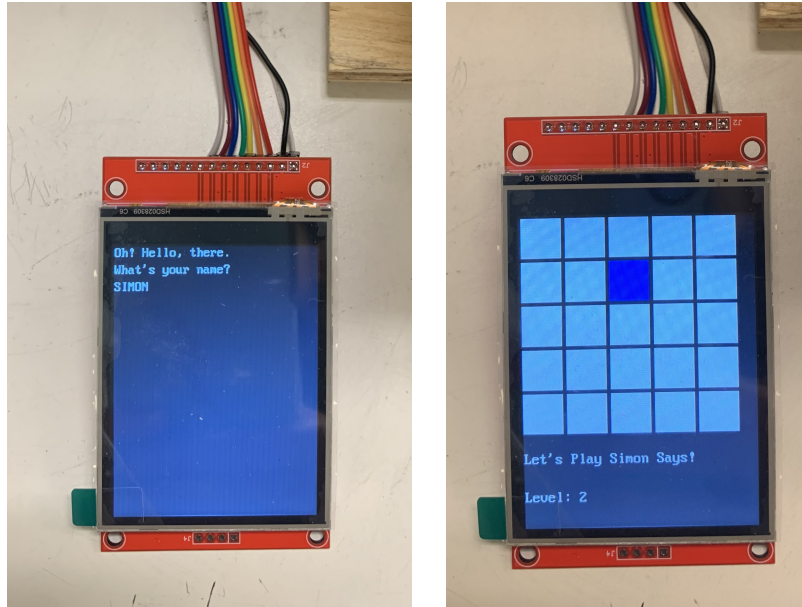


Figure 5: Example input name of "SIMON" to play Simon Says.

At the Code state, the user is prompted for a code which can be inputted using the capacitive touch sensor, where each copper strip corresponds to a 0-9 digit. The correct code is known by the user (in this case, it is 62442—yet another nod to Harry Potter). When inputted correctly, the door will be unlocked. Incorrect attempts will offer the user a new chance to input their name, instead.

4 PSoC Implementation

This project was implemented using multiple PSoC components to integrate the desired hardware.

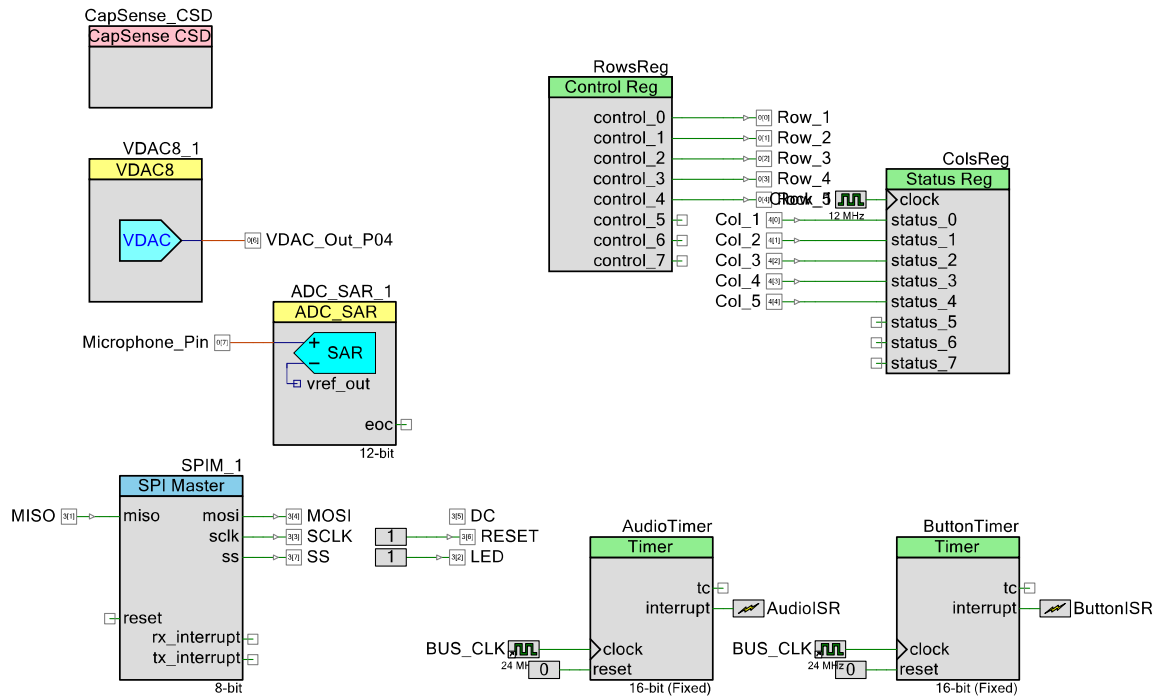


Figure 6: The PSoC TopDesign used in this project.

The figure above shows the different PSoC components used in this project:

- The CapSense_CSD component is used for the capacitive touch sensor.
- The control and status registers are used to get data from the button matrix, with the status of buttons being retrieved every time the ButtonTimer component sends an interrupt.
- The SPIM_1 is used to communicate with the TFT display.
- The speaker uses the VDAC8_1 to output proper audio whenever the AudioTimer sends an interrupt.
- The ADC_SAR_1 offers the option to also attach a microphone to this project.

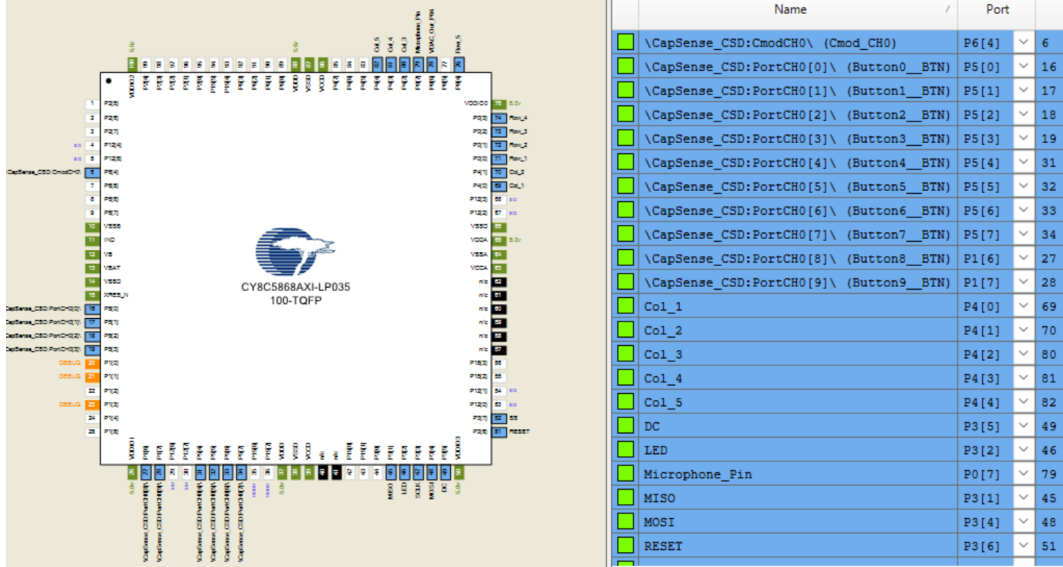


Figure 7: The PSoC pin mapping used in this project.

The figure above shows the different PSoC ports used in this project:

- Button Matrix: Rows $\rightarrow$ P0[0]-P0[4], Columns $\rightarrow$ P4[0]-P4[4]
- Capacitive Touch Sensor: CapSense CMOD $\rightarrow$ P6[4], Copper Strips $\rightarrow$ P5[0]-P5[7] and P1[6]-P1[7]
- TFT Display: MISO, LED, SCLK, SDI, DC, RESET, CS $\rightarrow$ P3[1]-P3[7]
- Speaker Input (i.e. VDAC Output): P0[6]

The full PSoC code can be found attached to the end of this report.

5 Final Design

The end result of this project is a functioning puzzle wall that allows the user to have a secret passageway within their home.

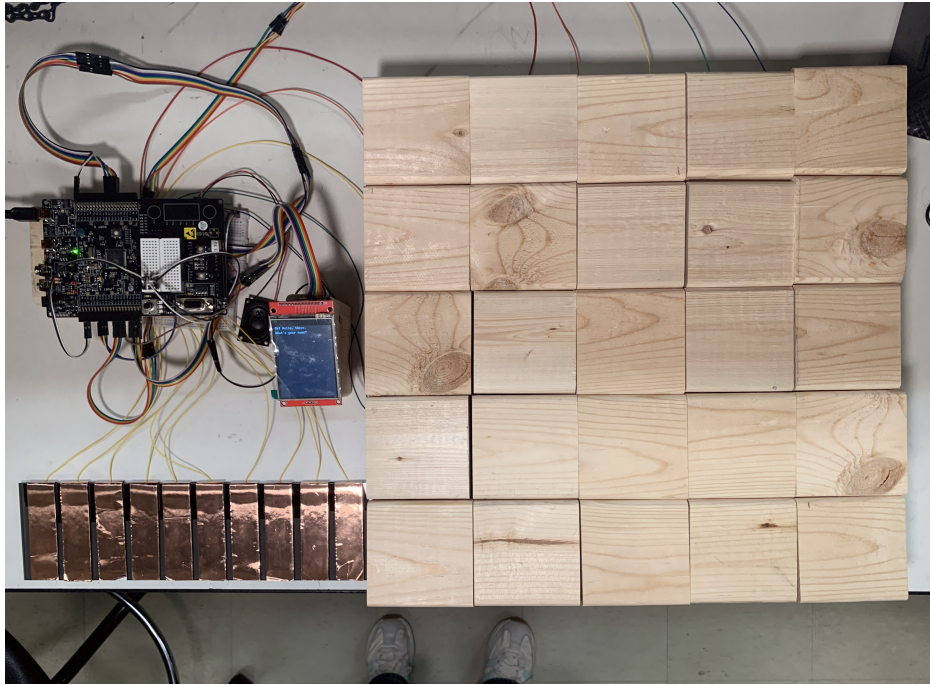


Figure 8: Final puzzle wall design!